

NARIN SRIPHO

Engineering Manager · Plant Engineering & Industrial Automation

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Mueang Samut Sakhon 74000, Thailand | Thai (native) & English (professional)

PROFESSIONAL PROFILE

Licensed Electrical Engineer and engineering leader who began on the shop floor designing injection and blow moulding tooling, and now runs plant engineering, heavy MEP infrastructure and factory automation as a single function. I bridge two disciplines that rarely sit under one leader: the **utility backbone** a plant runs on — solar, transformers, HVAC, fire protection, wastewater — and the **automation and data layer** above it, from PLC-driven conveyors to machine vision, IIoT and edge AI. What I build toward is a plant that decides from live data rather than paper, generates a growing share of its own energy, and builds its own automation instead of renting it from vendors.

404 kW SOLAR INSTALLED	1,500 kVA TRANSFORMER CAPACITY	100 / min SORTING LINE RATE	20+ yrs ENGINEERING & MGMT
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KEY ACHIEVEMENTS

quantified impact

฿2M PER YEAR · 33% CUT	Directed the end-to-end installation and commissioning of the plant's utility backbone — a 404 kW solar array, 1,500 kVA of transformers , fire protection and wastewater treatment. Cut plant energy costs by approx. ฿2,000,000/year (33%) and cleared every Thai regulatory inspection with zero compliance delays.
400% THROUGHPUT	Engineered a high-speed sorting line (100 items/min) with dual independent checkweighers, validated in 3D structural and kinematic simulation before fabrication. Raised sorting throughput 400% and avoided an estimated ฿1,000,000 in rework and structural modification.
80% CAPEX REDUCTION	Designed and deployed an in-house machine vision counting system (YOLOv8 on Raspberry Pi 5 with Coral Edge TPU) to replace manual tallying. Reduced capital expenditure by approx. 80% versus commercial vendor systems and eliminated 1 hour/day of manual tracking.
10 hrs PER WEEK RECOVERED	Built the plant's IIoT data layer end-to-end (n8n, Node-RED, Modbus RS485), migrating shop-floor sensors to live dashboards — retiring paper records and recovering 10 hours/week of administrative work.
฿20M+ CAPITAL PROGRAMME	Consolidated engineering, automation and IT into a single 12-person function and delivered a ฿20M+ capital programme — removing the vendor handoffs that delayed every plant-floor change and giving the plant one owner for energy, uptime and data.

01 Plant Utility Backbone			฿2M/yr saved
SOLAR · TRANSFORMERS · FIRE PROTECTION · WASTEWATER · WEIGHBRIDGE			
PROBLEM	APPROACH	RESULT	
The plant carried its full energy load from the grid and had no single owner for statutory utility infrastructure — every inspection cycle risked schedule slip.	Took delivery end-to-end: specification, contractor selection and supervision, regulatory inspection and commissioning through to handover across 404 kW of solar generation, 1,500 kVA of transformer capacity , HVAC, fire protection, wastewater and weighbridge.	Plant energy cost down approx. ฿2,000,000/year (33%) . Every Thai regulatory inspection cleared with zero compliance delays .	
Solar PV · MV/LV distribution · HVAC · Fire protection · Wastewater treatment · Contractor supervision · Commissioning			

02 High-Speed Sorting System			100 items/min
MULTI-STAGE CONVEYOR · DUAL CHECKWEIGHERS · SIEMENS S7-1200			
PROBLEM	APPROACH	RESULT	
Sorting throughput capped the line, and a mechanical redesign at this scale normally means discovering clearance and load faults on the shop floor — after fabrication.	Specified a multi-stage conveyor line with dual independent checkweighers running in parallel , driven by Siemens S7-1200 PLCs and pneumatic actuators — then validated the whole layout in 3D structural and kinematic simulation before cutting steel .	Throughput raised 400% to 100 items/min. Clearance and load issues caught at design stage, avoiding an estimated ฿1,000,000 in rework.	
SolidWorks · Kinematic simulation · Siemens S7-1200 / TIA Portal · Pneumatic actuators · Checkweigher integration			

03 Edge AI Vision Counter			80% lower capex
YOLOv8 · CORAL EDGE TPU · RASPBERRY PI 5 · CLOUD LOGGING			
PROBLEM	APPROACH	RESULT	
Product counting was manual — a full hour a day of tallying, with transcription error baked in. Commercial vision systems quoted at several times the budget.	Trained a YOLOv8 detection model via CiRA-CORE and Google Colab and deployed it on-line on Raspberry Pi 5 with a Google Coral Edge TPU , with results auto-logged to the cloud and surfaced on operator dashboards.	Capex cut roughly 80% versus vendor systems, 1 hour/day of manual tracking eliminated, and the counting logic kept in-house to retrain as products change.	
YOLOv8 · Google Coral Edge TPU · Raspberry Pi 5 · CiRA-CORE · Google Colab · NVIDIA RTX 5060 Ti			

04 Plant IIoT Data Layer**10 hrs/week**

n8n · NODE-RED · MODBUS RS485 · SELF-HOSTED VPS

PROBLEM

Shop-floor readings lived on paper and in spreadsheets — visible only after the fact, and only to whoever typed them up.

APPROACH

Wired IoT sensors into centralised databases over **Modbus RS485**, orchestrated with **n8n pipelines on a Docker/Portainer VPS**, and published live **Node-RED dashboards** with custom CSS/Angular front-ends for operators.

RESULT

Paper records retired and **10 hours/week** of administrative work recovered — the plant now reads its own state live rather than reconstructing it.

n8n · Node-RED · Modbus RS485 · Docker / Portainer · Webhooks & REST · Google Apps Script · Tailwind CSS

05 Non-Blocking Control Architecture**Real-time safe**

ESP32 · FreeRTOS · STATE MACHINE · FOOD PROCESSING

PROBLEM

Cloud logging on a food-processing controller competes with the control loop — a blocking transmit means a missed real-time deadline on live equipment.

APPROACH

Built the controller on **ESP32 + FreeRTOS** with data transmission isolated as a **separate background task**, applying state-machine design, ISRs and hardware debouncing; HMI panels integrated with relay boards over Modbus RS485 on an **isolated control LAN**.

RESULT

Cloud logging never interrupts real-time loops — the plant gets telemetry without trading away deterministic control.

ESP32 / ESP8266 · FreeRTOS · ISR & debouncing · State-machine design · Isolated control network

CORE COMPETENCIES**PLANT & HEAVY ENGINEERING**

MEP infrastructure delivery · Solar PV · Transformers & distribution · HVAC · Fire protection · Wastewater treatment · Weighbridge · Contractor supervision & commissioning

INDUSTRIAL AUTOMATION & CONTROLS

Siemens S7-1200 (TIA Portal) · PUT/GET remote communication · HMI-relay board integration via Modbus RS485 · Pneumatic actuator & conveyor control · Isolated control-network architecture

MACHINE VISION & EDGE AI

YOLOv8 object detection · Google Coral Edge TPU · Raspberry Pi 5 · Model training via CiRA-CORE / Google Colab · AnythingLLM enterprise knowledge base

MECHANICAL DESIGN & SIMULATION

SolidWorks · SketchUp Pro · 3D kinematic simulation for mechanical load, clearance and motion-constraint validation prior to fabrication

SAFETY, COMPLIANCE & LEADERSHIP

Safety Officer, Management Level (ฉบับบริหาร) · Statutory workplace-safety accountability · Cross-functional team & capital project delivery

TOOLING & PLASTICS PROCESSING

Injection & blow mould design · Mould construction supervision · Machine parameter setting & optimisation · Work Instruction (WI) authoring · New-model introduction to production

EMBEDDED & REAL-TIME SYSTEMS

ESP32 / ESP8266 · FreeRTOS multitasking · Non-blocking control loops · Interrupt service routines · Hardware debouncing · State-machine design for production equipment

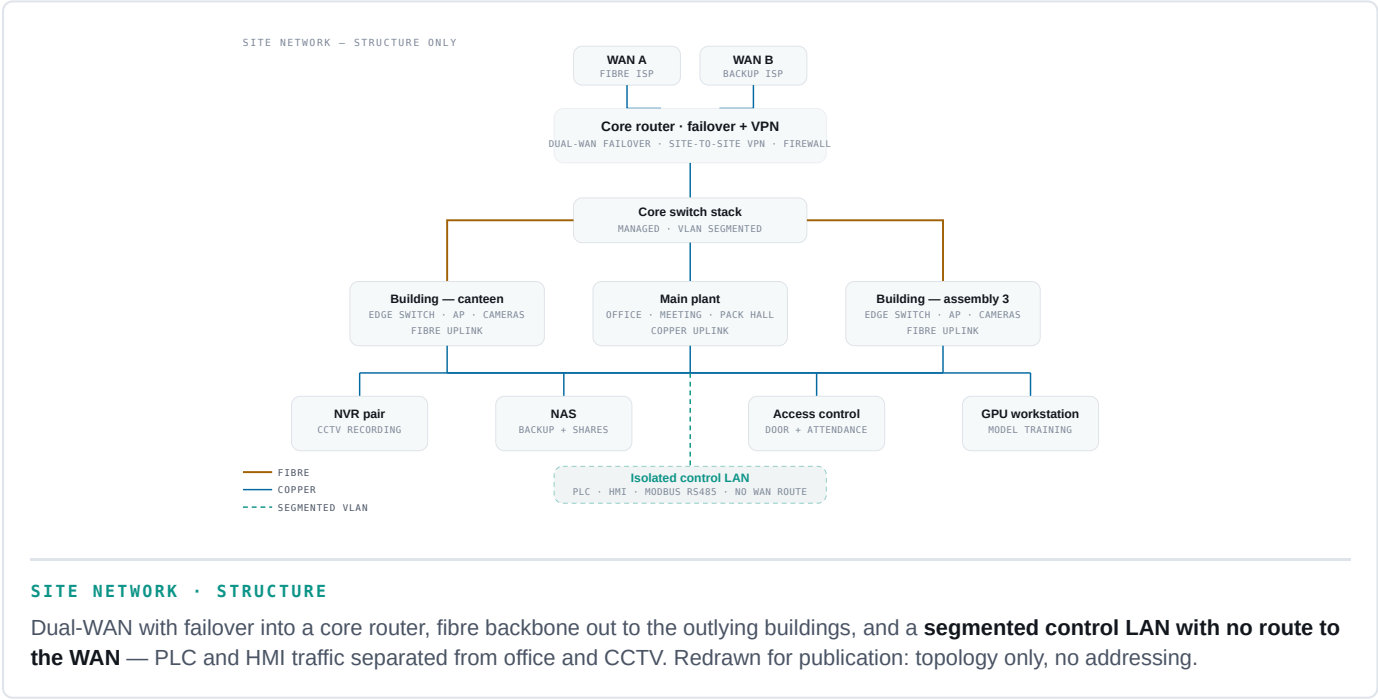
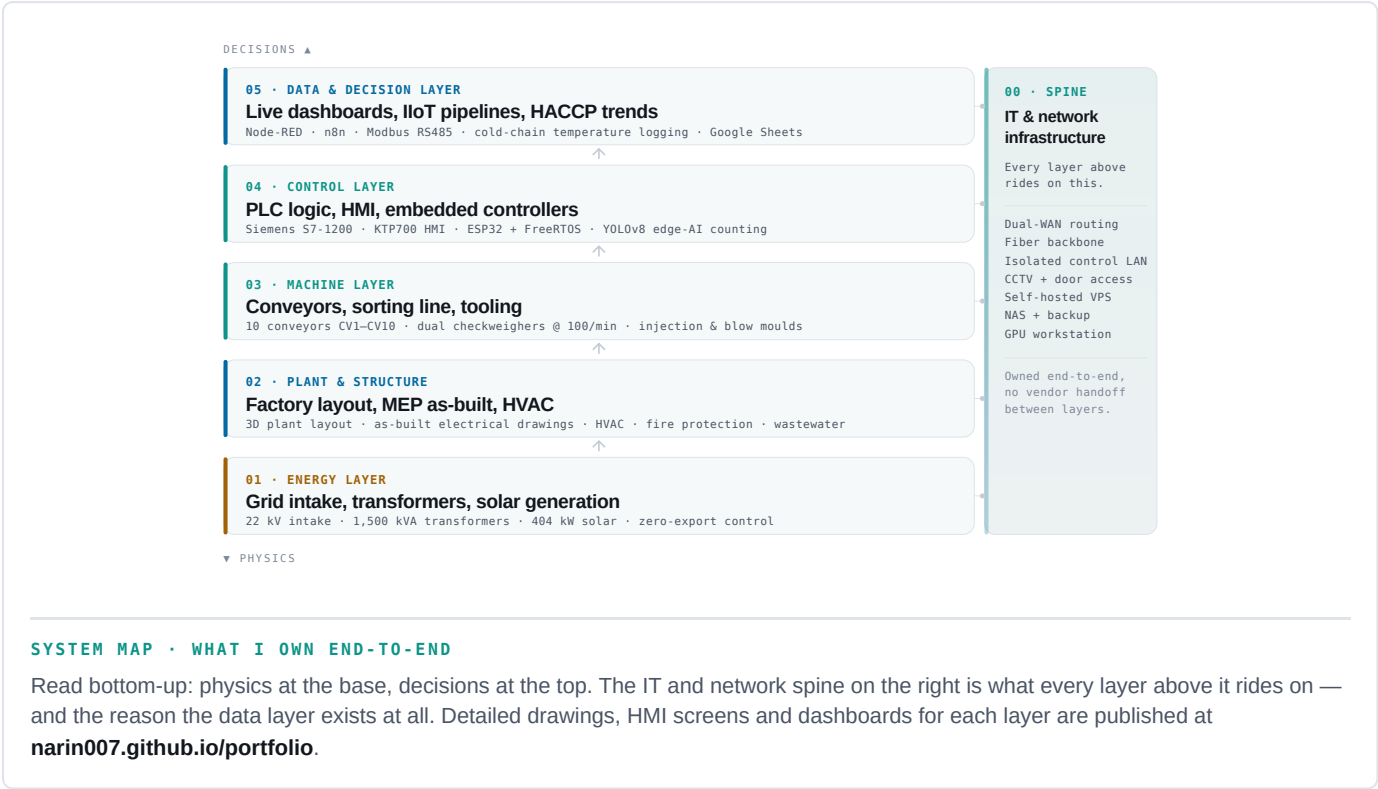
IIOT, DATA & APPLICATIONS

n8n pipelines on VPS (Docker / Portainer) · Node-RED dashboards (custom CSS/Angular) · Webhooks & REST · Google Apps Script · Tailwind CSS

IT & NETWORK INFRASTRUCTURE

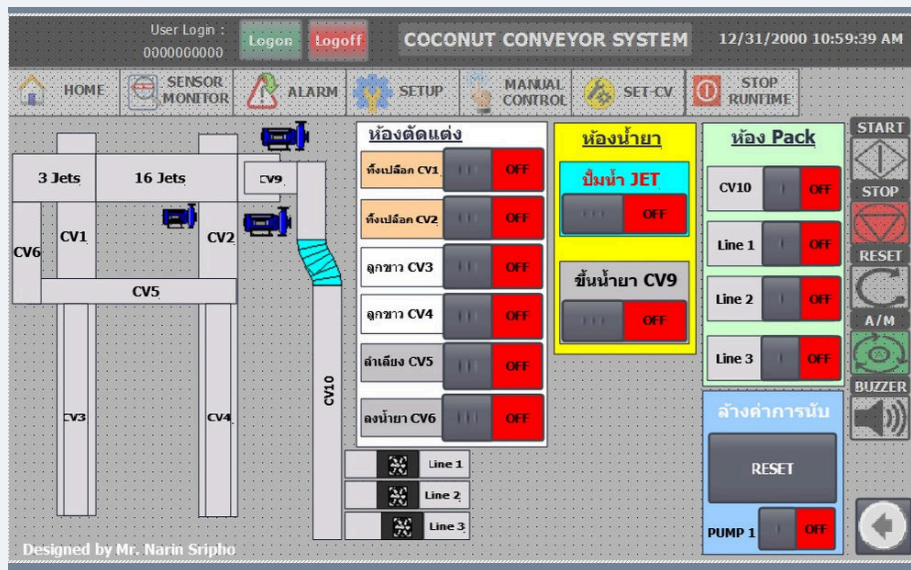
MikroTik routing · Windows SMB hardening · SSL domain proxy · CCTV & access control · GPU deployment (NVIDIA RTX 5060 Ti) · Legacy equipment management

A factory is five systems stacked on each other, and most plants buy each layer from a different vendor. At K-Fresh they sit under one function — which is why a change at the energy layer can reach the operator dashboard without a single handoff.



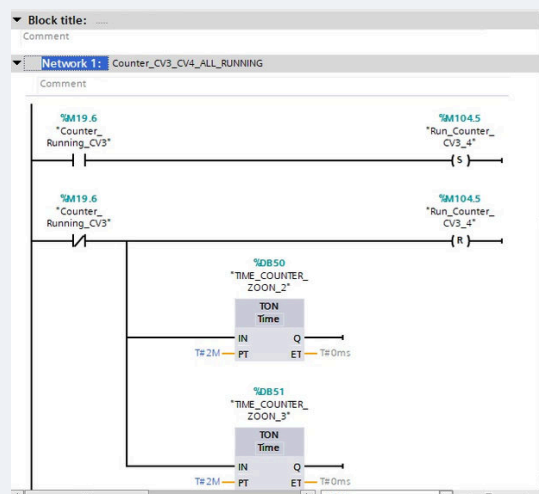
SITE NETWORK • STRUCTURE

Dual-WAN with failover into a core router, fibre backbone out to the outlying buildings, and a **segmented control LAN with no route to the WAN** — PLC and HMI traffic separated from office and CCTV. Redrawn for publication: topology only, no addressing.



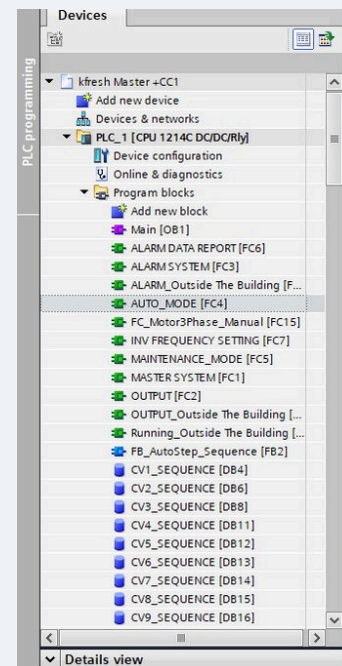
HMI DESIGN · SIEMENS KTP700 BASIC PN

Operator screen for the coconut conveyor line: **ten conveyors (CV1–CV10)** laid out as the plant actually sits, grouped by process area — trimming room, chemical room, pack room — with per-conveyor run state, three packing lines, jet-pump control and a counter reset. Navigation is split by task (Sensor Monitor / Alarm / Setup / Manual Control), and destructive actions are separated from routine ones. **Labelled in Thai for the operators who use it**, not in English for the engineer who wrote it.



LADDER LOGIC · AUTO_MODE [FC4]

Counter-running interlock built on a **set/reset coil pair** rather than a latched contact, so the running state has one owner and cannot be left asserted by a scan-order accident. Zone handoff is timed with **TON blocks on dedicated instance DBs** — each zone keeps its own timing state instead of sharing one timer.



PROGRAM STRUCTURE · CPU 1214C

Logic split into **named functions** — MASTER SYSTEM, AUTO_MODE, MAINTENANCE_MODE, ALARM SYSTEM, INV FREQUENCY SETTING, OUTPUT — with one **sequence DB per conveyor** (CV1–CV9). OB1 stays a call list, so a change to one conveyor is a change to one block.

ARCHITECTURE — 1 × HMI KTP700 Basic PN + 3 × CPU 1214C (master + two dedicated counting PLCs) on a shared PROFINET segment, with CM 1241 serial modules for RS485 devices. Counting load is kept off the master CPU so scan time stays predictable.

IT & Innovation Manager

SEPT 2020 – PRESENT

K-Fresh Co., Ltd. — Samut Sakhon, Thailand

PLANT INFRASTRUCTURE & HEAVY ENGINEERING

- Supervised end-to-end delivery of major plant infrastructure — **solar generation, transformers, HVAC, weighbridge, fire protection and wastewater treatment** — coordinating contractors, regulatory inspection and commissioning through to handover.
- Maintain and optimise heavy production machinery with **uptime accountability**, and carry statutory workplace-safety responsibility as certified Safety Officer at Management Level.

AUTOMATION, MACHINE DESIGN & CONTROLS

- Design and build mechanical conveyor systems integrated with **Siemens S7-1200 PLCs** and pneumatic actuators, taking concepts from 3D simulation through fabrication to production handover.
- Develop embedded controllers on **ESP32 + FreeRTOS** applying state-machine design, interrupt routines and hardware debouncing; integrate HMI panels with relay boards over Modbus RS485 on isolated control LANs.

MACHINE VISION, IIOT & IT INFRASTRUCTURE

- Deploy machine vision and edge AI for real-time product counting — training **YOLOv8** models and running them on Raspberry Pi 5 with Coral Edge TPU — with automated cloud logging and operator dashboards.
- Modernise plant workflows by wiring IoT sensors into centralised databases via **n8n and Node-RED**, replacing paper and spreadsheet reporting with live dashboards.
- Own corporate IT and security: CCTV, access control, MikroTik routing, Windows SMB hardening, Docker/Portainer on self-hosted VPS, and GPU workstations for model training.

Manager

MAR 2009 – AUG 2020

GTA Visa Expert Co., Ltd.

- Led general administration and internal organisational operations across an **11-year tenure** — accountable for process design, staffing and day-to-day management, building the operational foundation later applied to running a plant engineering function.

Engineering — New Model

2005 – 2007

Supawut Co., Ltd. (บริษัท สุภาวุฒิ จำกัด)

- Designed **injection and blow moulding tooling** for new-model parts, owning the design from concept through to a production-ready mould.
- Supervised mould construction, verifying dimensional accuracy and resolving design issues before trial runs to avoid rework at the machine.
- Set and optimised process parameters on injection and blow moulding machines to bring new tooling to stable, repeatable output.
- Authored **Work Instructions (WI)** defining process settings and quality checks, then handed the validated process over to Production for volume manufacture.

EDUCATION & CREDENTIALS**DEGREE****B.Eng. Electrical Engineering
(Telecommunications)**

Rajamangala University of Technology
Thanyaburi (RMUTT)
Graduated 2005 · GPA 3.27

PROFESSIONAL LICENSE**Associate Engineer License,
Electrical**

Council of Engineers Thailand · status
active

License No. กฟท. 68797**STATUTORY CERTIFICATION****Safety Officer, Management Level
(อป.บริหาร)**

Certified under Thai Ministry of Labour
occupational safety regulations

AVAILABILITY

Open to **Engineering Manager, Plant Engineering Manager** and **Automation / Industrial Digitalisation** leadership roles — permanent or contract, Thailand-based with regional travel.

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Samut Sakhon

LOCATION